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ORIGINAL ARTICLE

Ozone Trends and Mortality Risk: The Growing Need for Machine Learning Predictions in Bogotá, Colombia

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Abstract

Tropospheric ozone has the potential to become an increasingly pressing public health issue in Bogotá, Colombia, due to rising concentrations across the city driven by complex interactions among emissions, meteorology, and urban structure. This study presents a comprehensive spatiotemporal analysis of ozone levels from 2013 to 2023 and assesses the associated health burden using mortality data from the same period. Results reveal a consistent upward trend in ozone concentrations, particularly in northern, western, and southern localities, with seasonal peaks linked to biomass burning and photochemical conditions. Mortality analysis, based on the Global Exposure Mortality Model, estimates that 18.3% of all deaths among individuals aged 25 and older are attributable to long-term ozone exposure. The highest burdens are found in densely populated and socioeconomically vulnerable areas such as Kennedy, Suba, and Ciudad Bolívar, with the elderly being the most affected. Building on these findings, we developed a machine learning prediction model for ozone using a convolutional merge with a long-short term memory network architecture trained on air quality and meteorological variables. The model demonstrated strong predictive performance (mean Rho=0.86, RMSE=3.5 $\mu g/m^3$) across monitoring stations (17 with at least 35000 data points), supporting its potential application in real-time early warning systems across Bogotá. This integrated approach highlights the importance of localized air quality management, combining epidemiological assessment with predictive modeling. The findings underscore the urgency of implementing region-specific mitigation strategies and improving monitoring infrastructure to reduce health risks from ozone exposure in Bogotá's rapidly growing urban environment.

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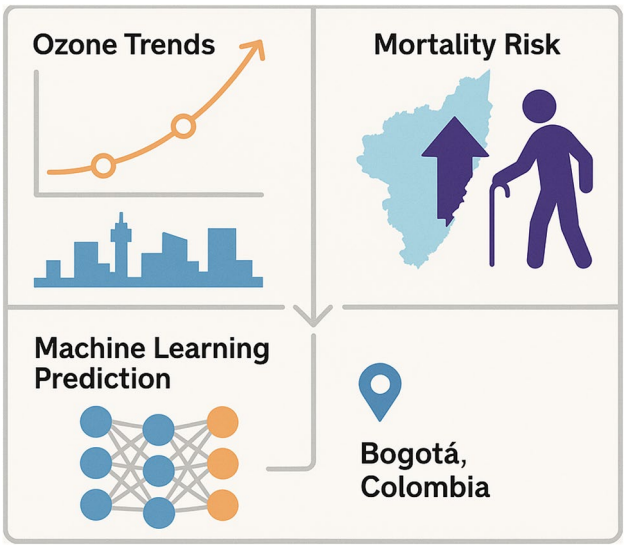
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Graphical Abstract



The top-left panel shows a rising trend in tropospheric ozone concentrations across Bogotá, highlighting the increasing severity of air pollution. The top-right panel represents the associated health burden, indicating elevated mortality risks, particularly among older and vulnerable populations. The bottom-left panel depicts the machine learning framework developed in this study, integrating a convolutional layer and a bidirectional long-short term memory architecture to predict future ozone levels with high accuracy, showing its potential as a base for early warning systems. Finally, the bottom-right panel emphasizes the study's geographical focus on Bogotá, Colombia.

Highlights

- Ozone concentrations in Bogotá have shown a significant upward trend (2013–2023).
- Long-term exposure to ozone is associated with a 18.3% increase in mortality risk among individuals aged 25 and older.
- The highest mortality burdens are concentrated in socioeconomically vulnerable localities.
- A machine learning model demonstrates strong potential for real-time ozone prediction and early warning systems.

Keywords Tropospheric ozone · Air pollution · Mortality burden · Machine learning · Public health · Development of a prediction model

1 Introduction

Atmospheric pollution, driven by both natural processes and human activities, has become one of our planet's most urgent health challenges. In 2019, it ranked as the fourth leading cause of death worldwide, with some 4.2 million lives lost to heart disease, stroke, respiratory infections, and other air pollution-related illnesses (WHO 2021). The burning of fossil fuels for power generation, transport (Lelieveld et al. 2023), and industrial processes remains the primary anthropogenic source of harmful emissions (WHO 2021).

Among the array of pollutants, tropospheric ozone (O_3) poses a particular threat to human health and ecosystems. Unlike primary pollutants, O_3 is formed by photochemical reactions involving nitrogen oxides (NO_x) and volatile organic compounds (VOCs), so efforts to curb NO_x emissions, while successful at reducing nitric oxides, can sometimes lead to unintended rises in ground-level ozone (Sokhi et al. 2022). Globally, O_3 levels have shown a persistent upward trend since the 1990s, particularly in rapidly urbanizing regions and areas with increasing background pollution (Chang et al. 2017; Malashock et al. 2022). In Latin America, megacities such as Mexico City and São Paulo have reported similar challenges with rising ozone levels despite reductions in primary pollutants, highlighting the complexity of secondary pollutant control (Seguel et al. 2024). Satellite observations and surface stations alike have tracked these dynamics globally (Monks et al. 2015), and in Colombia the SISAIRE air quality information system (IDEAM 2023) has underpinned numerous studies of local O_3 impacts, trends, and drivers (Ballesteros et al. 2020; Henao et al. 2021).

Recent research highlights two key drivers of elevated ozone in Colombian cities. First, biomass burning during

the dry season, whether wildfires along the Venezuelan border or regional agricultural fires, principally produce in the Colombian Amazon, especially in Caquetá (Celis et al. 2023), injects both ozone precursors and particulate matter, which are then transported by prevailing winds into urban basins (Bolaño-Diaz et al. 2022; Casallas et al. 2023). For example, in Bogotá, March and April register the highest seasonal peaks, coinciding with both biomass-burning and meteorological conditions that favor pollutant accumulation (Casallas et al. 2024). The second driver is an unexpected ozone surge produced during the COVID-19 lockdowns (~70% in Bogotá and ~30% in Medellín), which reveals the non-linear chemistry of urban photochemistry (Sokhi et al. 2022). In VOC-limited environments like these cities, steep NO_x cuts can reduce the titration reaction ($NO + O_3 \rightarrow NO_2 + O_2$), so less NO scavenges less O_3, and ambient ozone climbs (Seinfeld And Pandis 2016; An et al. 2024; Yin et al. 2025). In essence, while cutting NO_x is critical to curb particulate and primary gaseous pollutants, doing so in isolation may backfire on ozone levels unless VOCs are simultaneously controlled.

These lockdown findings carry a clear policy lesson: air-quality strategies must be multi-pollutant and regionally tuned. Fuel-efficiency standards and cleaner vehicle technologies remain essential, but by themselves they risk shifting the chemical regime toward higher ozone if VOC emissions (from solvents, industry, biomass burning, or even biogenic sources) are not also targeted. Effective ozone mitigation thus demands coordinated measures: tighter VOC controls (e.g., on industrial solvents and evaporative losses), continued NO_x reductions, and real-time photochemical modeling to guide adaptive interventions (Sokhi et al. 2022; WMO 2023). Recent machine-learning studies have demonstrated the value of such approaches by improving spatiotemporal ozone prediction, characterizing nonlinear ozone-formation sensitivities, and strengthening ozone exposure–health risk assessments across diverse chemical regimes (e.g., Ma et al. 2024; Hosseinpour et al. 2024). As shown in countries that explicitly integrate tropospheric ozone modeling into air-quality management frameworks, embedding ozone-formation sensitivities into transport and fuel policies is essential to ensure that gains in one pollutant do not translate into unintended setbacks for another.

Building on these previous studies, an important question arises: in recent years, with the reduction of NO emissions due to advances in fuel quality and technology, has ozone increased, particularly in Colombian cities, which appear to be especially sensitive to such changes? To answer this, Casallas et al. (2024) investigated ozone trends between 2017 and 2022 at a regional level and reported a significant increase in O_3 levels in Bogotá. This increase appears to be linked to reduced atmospheric titration (coinciding with the lockdown results, although the increase lasts longer than the lockdowns) resulting from a decline in NO emissions. Importantly, their analysis also identified temperature as a significant factor influencing ozone variability, suggesting that meteorological changes play a growing role in shaping urban ozone dynamics. Although current O_3 concentrations remain below the maximum permissible limit established by Resolution 2254 of 2017 (Ministry of Environment And Sustainable Development 2017), the observed upward trend signals a potential future concern.

This aforementioned temperature sensitivity aligns with broader concerns about climate change (Zeng et al. 2022). Rising global temperatures accelerate the photochemical reactions that produce ozone and increase the frequency of stagnation events, which trap pollutants near the surface (Yim et al. 2019). Additionally, urban heat island effects (caused by increased land surface temperatures in built-up areas) can amplify local ozone formation (Song et al. 2025; Wang et al. 2025). Together, these climate-related factors (Ban et al. 2022) are likely to exacerbate ozone pollution in fast-growing cities like Bogotá, underscoring the importance of continued monitoring, deeper understanding of ozone dynamics, and the implementation of early warning systems to safeguard public health (WMO 2023).

Motivated by these concerns, the present study focuses on three main objectives: first, to determine whether the observed increase in ozone levels in Bogotá is a regional phenomenon or driven by localized urban factors; second, to assess the impacts of ozone on public health using mortality data; and third, to develop a predictive model for ozone that can serve as a foundation for an early warning system.

2 Data and Methods

2.1 Study Area

Bogotá, the capital of Colombia, is located in the center of the country, forming part of the Andean region (Fig. 1). It sits on a plateau at around 2600 meters above sea level (masl) on the eastern branch of the Andes mountain range and is surrounded by a system of mountains that reach an approximate altitude of 3100 masl. Additionally, the extremely complex topography of the Andes creates particularly high spatial variability in its meteorological characteristics, especially in terms of boundary layer height, precipitation, and wind properties (Hernández-Deckers et al. 2021), and have a big impact on the evolution of pollutant emission (Antezana-Lopez et al. 2025). The strategic location of the city means that Bogotá does not have distinct climatic seasons and experiences a homogeneous climate throughout the year with an average temperature of 14 °C (District Department

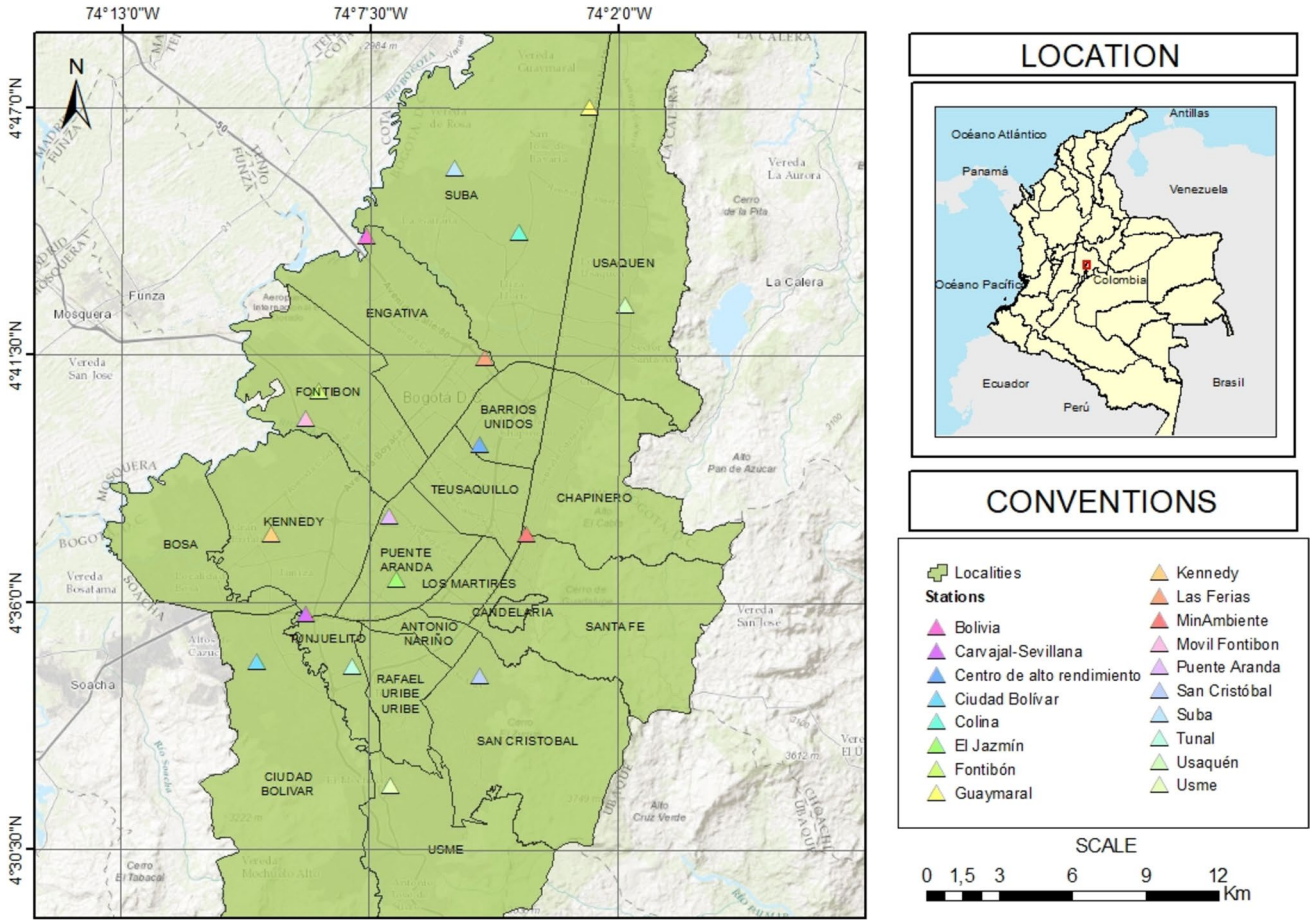


Fig. 1 Location of the study area in Bogotá, including the city's localities and the air quality monitoring stations of the RMCAB (markers)

of Environment 2023). The city has a bimodal rainfall pattern, typical of the northeastern Andean region, with two dry seasons and two periods of high precipitation. The rainiest periods are April–May, and October–November (Guzmán et al. 2014).

Bogotá has a well-established air quality monitoring network (RMCAB, by its Spanish acronym). Using this network, influential variables affecting the behavior of O_3 between 2013 and 2023 were analyzed. The RMCAB network is now operating 18 air quality stations. Table SM1 in the Supplementary Material summarizes the monitoring stations, including their temporal resolution, measured pollutants, and data coverage period.

2.2 Temporal Trends in Ozone Concentration

Here we performed a comprehensive spatiotemporal analysis of O_3 levels across different areas of Bogotá. This includes evaluating long-term trends and identifying the key factors influencing ozone concentrations. Before using the data, Chauvenet's criterion is applied, following the Lishu et al. (2015) approach to reduce uncertainty. Here, we adapt the criterion, traditionally used with parametric distributions, to flag values exceeding three standard deviations above the mean. These flagged values are then cross-checked against observations from the three closest surface monitoring stations to determine whether they reflect regional phenomena or are likely due to sensor anomalies.

We selected Chauvenet's criterion based on its effectiveness in identifying extreme deviations in environmental time series while preserving the statistical properties of the dataset. To assess its robustness, we compared it with other commonly used outlier detection methods such as the Interquartile Range (IQR), Median Absolute Deviation (MAD), and Level Shift Detection, using the Anomaly Detection Toolkit (ADTK; Zhao et al. 2020). The results were nearly identical across these methods, with flagged outliers differing by less than 3% overall. This supports the validity of our choice while maintaining methodological consistency. It is important to remark that no imputation or interpolation was applied to missing data at this stage; missing values were only addressed during the machine learning preprocessing step, as described in the relevant section.

To analyze spatial and temporal trends in ozone concentrations across Bogotá, we used daily data of the maximum 8-hour running average (hereafter OSMDA8) to calculate multi-year averages at key monitoring stations. Then, we applied the Seasonal-Trend decomposition using LOESS (STL) method developed by Cleveland et al. (1990), which uses Locally Weighted Scatterplot Smoothing (LOESS), a linear regression technique that smooths time series by fitting localized, weighted regressions. To assess long-term changes, we smoothed the data and normalized its distribution using STL, then calculated the monthly trend slope using Generalized Least Squares (GLS) regression with a 95% confidence interval. GLS is appropriate for modeling autocorrelated time series, allowing for the estimation of independent residuals and accurate regression coefficient variances (Brockwell And Davis 2002; Singh et al. 2021). The model used for GLS estimation is described in Eq. 1.

$$yt = Xt\beta + \epsilon t \tag{1}$$

Where yt is the dependent variable at time t, Xt is a vector of independent variables, β represents the coefficients to be estimated, and ϵt denotes the error term at a specific time.

2.3 Annual Mortality Associated with Tropospheric Ozone

To complement the ozone trend analysis, it is also important to assess the potential public health implications of long-term exposure. To this end, mortality data provided by the District Health Secretariat of Bogotá for the period 2013–2023 were analyzed (DANE 2019). The focus was on mortality outcomes known to be strongly associated with ozone exposure, including all-cause mortality (A00–Z99) and cause-specific categories such as circulatory diseases (I00–I99), ischemic heart diseases (IHD, I20–I25), cerebrovascular diseases (CVA, I60–I69), respiratory diseases (J00–J99), chronic obstructive pulmonary diseases (COPD, J40, J44, J47), acute lower respiratory infections (ALRI, J12–J18, J20–J22), and lung cancer (C30–C39). Additionally, mortality from natural causes (A00–R99), which excludes external causes such as accidents, suicides, and homicides, was included to better isolate health outcomes potentially linked to environmental factors like air pollution. Furthermore, the population estimates by municipality and 5-year age groups were obtained from DANE based on the national census 2018 (DANE 2024).

To estimate the health risks associated with projected ozone concentrations, two established methodologies were employed: The Global Exposure Mortality Model (GEMM) developed by Burnett et al. (2018) and the Exposure-Response Function (ERF) approach proposed by Bell et al. (2005). GEMM is based on the use of hazard ratios and enables the standardization of health impact assessments across different populations and studies. The ERF is a mathematical model that quantifies the relationship between the level of exposure to a pollutant and the associated risk of a specific health outcome. This function is essential for estimating mortality attributable to air pollution, as it captures how health risks vary with different levels of exposure.

Since these illnesses can be caused by multiple factors beyond ozone exposure, confounding control was essential (Bell et al. 2005; Burnett et al. 2018). In the ERF model, we adopted the framework established by Bell et al. (2005), which includes adjustments for day-of-week effects, long-term trends, and seasonality using a natural cubic spline of time (7 degrees of freedom [df] per year). Additionally, nonlinear meteorological effects were controlled using spline terms for same-day temperature (6 df), 1–3 day lagged mean temperature (6 df), same-day dew point (3 df), and 1–3 day lagged dew point (3 df). These components are widely used in environmental epidemiology and are effective in isolating the short-term effects of ozone by accounting for major time-varying confounders (Bell et al. 2005).

Although individual-level factors such as smoking and socioeconomic status (SES) are known mortality risk factors, they vary slowly over time and are not expected to confound short-term time-series associations. The time-series framework inherently controls for such time-invariant or slowly-varying population characteristics. Furthermore, for long-term exposure assessments using GEMM, the underlying hazard ratios were obtained from multi-cohort studies that explicitly adjusted for age, sex, smoking, and SES (Burnett et al. 2018), allowing us to incorporate those corrections into our estimates by proxy.

2.4 GEMM - Global Exposure Mortality Model

The GEMM is used to analyze a broad group of causes of mortality, including all noncommunicable diseases and lower respiratory infections (Burnett et al. 2018), in which the Eqs. 2 and 3 are used

$$HR(z) = \exp(\theta T(z) w(z)) \tag{2}$$

Where θ is an adjustable parameter, z is equal to $C - a$ with C being the ozone concentration level and a being the $\min(C)$. $T(z) = \ln(z + 1)$, and $w(z)$ is a function used to control the shape of the distribution curve. In the general case, we can use Eq. 3, but the expression depends on the distribution of the data.

$$w(z) = \frac{1}{1 + \exp(-\frac{z-\mu}{v})} \tag{3}$$

Where μ and v are parameters to be adjusted depending on the characteristics of the city (Burnett et al. 2014, 2018). The GEMM is calculated through a series of steps, beginning with data collection on the concentration of the risk factor, mortality rate, and exposed population. The data is then adjusted to account for variables that may influence mortality, such as age, sex, smoking, and socioeconomic status. Next, the relative risk of dying from a specific cause is estimated by comparing individuals exposed to the risk factor with those who are not. Finally, the impact on mortality is determined by multiplying the relative risk by the number of exposed individuals (Pozzer et al. 2023).

2.5 Calculation of Premature Deaths Attributable to Ozone Using Exposure-Response Function - ERF

To analyze the relationship between ozone exposure and mortality risk, the Poisson regression model used by Bell et al. (2005) is applied, as shown in Eq. 4

$$\begin{aligned}\log\left(\mu_t^c\right) =& \beta_c^t x_t^c + y^c DOW_t + ns\left(time_t, 7/year\right) + ns\left(T_t^c, 6\right) \\ &+ ns\left(T_{t-1,t-3}^c, 6\right) + ns\left(D_t^c, 3\right) + ns\left(D_{t-1,t-3}^c, 3\right) + I\end{aligned} \tag{4}$$

Where, μ_t^c represents the expected daily mortality rate in the community c on day t. β_t^c is the community-specific coefficient used to estimate the effect of ozone exposure on mortality, while x_t^c denotes the measure of ozone exposure in the community c on day t, defined as the average ozone concentration for the current day and the previous day (lag 0-1). $y^c DOW_t$ represents the day-of-week effects in the community c on day t. The term $ns(time_t, 7/year)$ is a natural cubic spline function of time with 7 degrees of freedom per year, included to adjust for long-term trends and seasonal variations.

$ns\left(T_t, 6\right)$ is a natural spline function of current-day temperature in the community c with 6 degrees of freedom, accounting for non-linear temperature effects. $ns\left(T_{t-1,t-3}^c, 6\right)$ represents a spline of the average temperature over the previous three days, also with 6 degrees of freedom. $ns\left(D_t^c, 3\right)$ is the natural spline function of dew point temperature on day t with 3 degrees of freedom, while $ns\left(D_{t-1,t-3}^c, 3\right)$ captures the average dew point over the previous three days (adjusted for the current day), also with 3 degrees of freedom.

I represents the interaction terms for age and time and refers to the interactions between natural spline functions of time and age-specific indicators (<65, 65-74, and $\geq$ 75 years), allowing the model to capture age-differentiated temporal trends. The β-value, used to estimate preventable deaths from diseases attributable to ozone exposure, is presented in Table 1, which is calculated using multi-year averages of ozone concentration and mortality rates by cause. It does not assess year-on-year variation but rather the total or average long-term attributable burden.

2.6 Machine Learning Method

2.6.1 Input Data and Variable Selection

To investigate the factors influencing ozone behavior, input variables were selected based on a review of scientific literature identifying key contributors to ozone formation in urban environments. These include emissions of ozone precursors, meteorological conditions that affect pollutant transport between regions (Liu et al. 2021), and atmospheric chemistry and biogenic processes (Wu et al. 2020), all of which are critical for predicting changes in ozone levels. Based on previous literature, ten variables were selected for analysis: O_3, CO, NO, NO_2, $PM_{2.5}$, PM_{10}, solar radiation, temperature, wind direction, and wind speed, coinciding with the variables selected by Li et al. (2019) and Casallas et al. (2023).

It is important to note that the objective of this study was to develop a high-performing predictive model suitable for operational early warning applications rather than to explore feature interpretability. Therefore, we did not conduct a new feature selection analysis or an interpretability study. The input variables were selected based on the comprehensive work of Li et al. (2019), who identified the main drivers of ozone dynamics in urban environments, and Casallas et al. (2023), who applied a Variance Inflation Factor (VIF) procedure for the Bogotá region to reduce multicollinearity and confirm the relevance of the selected predictors. The latter study, focused specifically on feature selection and interpretability, concluded that the NO to NO_2 ratio and its evolution, and the presence of wildfires are the dominant determinants of ozone variability in Bogotá. Building upon these findings, our study focuses on model accuracy and operational applicability for real-time ozone forecasting.

CO and NO_x were selected as ozone precursors due to their photochemical reactions. Although their relationship is not linear, it is essential to study precursors such as NO,

Table 1 β-value used for the endpoint of avoidable deaths for diseases due to exposure to ozone (Bell et al. 2005)

Endpoint Case	RR (95%; CI)	$\beta\mathbf{O_3}$ (Bell et al. 2005)
All-cause mortality	1.0087 (1.0055; 1.0117)	0.87 ± 0.11 %/10 ppb
Cardiovascular mortality	1.0110 (1.0068; 1.0152)	0.110 ± 0.10 %/10 ppb
Respiratory mortality	1.0047 (-1.0050; 1.0146)	0.47 ± 0.2 %/ 10 ppb

which can consume ozone due to the reduction in O_3 titration and radicals (Liu et al. 2020). $PM_{2.5}$ was included as they indicate transported pollution and organic aerosols, largely derived from biogenic emissions of VOCs. Although their impact is less than that of anthropogenic emissions, they can still affect O_3 formation (Wu et al. 2020). Regarding PM_{10}, it directly influences ozone formation by reflecting solar radiation and dilating the photochemical reaction. Therefore, in the absence of PM, O_3 formation increases (Ivatt et al. 2022). Additionally, PM can serve as a pr for wildfires, which, as Ballesteros et al. (2020) showed, are very important sources of ozone in Bogotá.

Meteorological variables were also analyzed in the formation of O_3, such as atmospheric humidity, temperature, solar radiation, precipitation, wind direction and speed, and the intrusion of stratospheric air masses (Monks et al. 2015). Meteorological patterns coincide with anomalies in O_3 levels and the correlation between these variables. For example, higher radiation, lower humidity, and higher-pressure lead to an increase in O_3 (Deroubaix et al. 2021). Once the variables were selected due to their potential relation with ozone, a VIF method was applied to exclude variables that have a multicollinearity above 5 (Kline 2015) and can lead to errors in the model (Ghahremanloo et al. 2021).

The model was trained and evaluated using data from the RMCAB, excluding stations with incomplete or insufficient data, such as Bosa, Colina, Móvil Fontibón, and Bolivia. These stations were excluded either due to a short period of ozone monitoring or a complete lack of relevant measurements. The remaining stations provided robust and continuous datasets suitable for model development. For the trained stations, missing values were imputed using the station-specific hourly mean, ensuring temporal consistency without introducing artificial trends (Celis et al. 2022). We also tested an alternative imputation approach using linear regression and dense neural networks, incorporating data from the nearest fully operational stations. However, simulation results showed no substantial differences between this advanced method and the mean-based imputation, and we therefore retained the simpler approach for consistency and computational efficiency.

It is important to mention that a key limitation of this study is the absence of certain known ozone precursors; (particularly VOCs and non-methane hydrocarbons [NMHCs]) in the input dataset. These compounds play a well-documented role in ozone formation, especially in urban environments. However, they are not currently measured by Bogotá's air quality monitoring network, as indicated in Table SM1. Although previous studies (e.g., Casallas et al. 2023) have incorporated satellite-derived VOC estimates from products, the mismatch in temporal resolution and limited improvement in model accuracy made their inclusion less suitable for our objective. Since our model is intended for real-time operational deployment, we prioritized variables that are consistently and directly measure by ground-based stations to ensure practicality and reproducibility.

2.6.2 Neural Network Model Structure and Validation

To forecast O_3 levels in Bogotá, we developed a machine learning model based on the framework proposed by Celis et al. (2022). The model architecture combines 1D convolutional layer (CNNs) with bidirectional Long Short-Term Memory (BiLSTM) layers. The model was implemented using Python libraries TensorFlow (Abadi et al. 2015), Keras (Chollet 2015), and Scikit-learn (Pedregosa et al. 2011). Specifically, we evaluated more than ten candidate architectures using a grid-search approach, varying (removing) the number of convolutional and recurrent layers, filter sizes, neuron counts, and regularization parameters. The architecture presented here was chosen because it consistently achieved the best performance on the validation dataset across three different evaluation metrics.

The architecture includes three 1D convolutional layers with 16, 32, and 64 filters, respectively, using a kernel size of 3 and the tanh activation function (Fukushima 1969) to introduce non-linearity. These layers identify local temporal patterns in the input data. A batch normalization layer follows to stabilize and accelerate the learning process. The output then passes through two BiLSTM layers, with 32 and 16 neurons, respectively, and a 20% dropout rate to prevent overfitting. The network concludes with three dense layers (32, 32, and 24 neurons), also using ReLU activation (Do et al. 2023). An early stopping method is used to avoid overfitting (Prechelt 1998).

Model training employed the Adam optimizer (Kingma And Ba 2017). Several loss functions were tested; Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), with Maximum Likelihood Sequence Estimation (MLSE) ultimately selected for its superior performance in capturing small peaks and preserving the shape of ozone predictions. Hyperparameters, including the number of neurons, dropout rates, and activation functions, were optimized via grid search. The training was performed over 50 epochs with a batch size of 64, followed by a final fitting step using 1 epoch and a batch size of 10. Notably, the model was trained on 24-hour lagged input data to enable forecasting of future ozone concentrations.

The trained model was validated using real-world data from Bogotá's air quality monitoring network. Validation followed the methodology of Celis et al. (2022), using 182 days of data selected across a range of Air Quality Index categories. This approach ensured the model was evaluated

under varying pollution scenarios, from clean to highly polluted days.

Six statistical and three categorical performance metrics were calculated to assess the model, as in Celis et al. (2022). Statistical metrics included Spearman's rank correlation (Rho), Root Mean Square Error (RMSE), Index of Agreement (IOA), Mean Bias (MB), Normalized Mean Bias (NMB), and Factor of Two (FAC2). The categorical metrics included the Hit Rate (HIT)—the proportion of true exceedances that are correctly predicted; the False Alarm Rate (FAR)—the proportion of false exceedances predicted; and the Proportion of Correctness (POC)—the percentage of predictions falling within acceptable thresholds. Categorical thresholds were based on Colombian air quality standards and WHO guidelines, with 100 ppb over an 8-hour average used as the exceedance limit, given the significant diurnal variability in ozone concentrations (WHO 2021). This threshold aligns the model output with regulatory and public health action criteria. All the statistical and categorical parameters were compared with benchmarks taken from previous studies (Emery And Yarwood 2001; Casallas et al. 2024), to account for model precision.

In addition to the main CNN–BiLSTM model, we evaluated several baseline models for comparison, including Random Forest (RF), LASSO regression, and a Support Vector Machine (SVM). The average Spearman correlation coefficient across all stations was 0.834 for RF, 0.716 for LASSO, and 0.762 for SVM, whereas the CNN–BiLSTM model achieved the highest correlation, with an average value of 0.863. Consistent improvements were also observed for error-based metrics. The NMB was substantially reduced for the CNN–BiLSTM model (6%) compared with RF (13%), SVM (27%), and LASSO (55%). Similarly, the CNN–BiLSTM model yielded the lowest root mean square error (RMSE = 1.75), outperforming RF (RMSE = 3.3), SVM (RMSE = 4.4), and LASSO (RMSE = 7.8). Taken together, these results demonstrate the superior ability of the CNN–BiLSTM architecture to capture the nonlinear and temporal dynamics of ozone concentrations in Bogotá.

3 Results and Discussion

3.1 Spatial Analysis of Ozone in Bogotá

Using data from the RMCAB, we perform a spatial analysis of O_3 concentrations in Bogotá, based on the multi-year mean calculated from the OSMDA8. Figure 2 shows that the ozone has significant patterns of variation, probably due to factors such as mobility restrictions, fuel quality, and meteorological anomalies. In 2020, elevated concentrations exceeding the safety thresholds (>25 µg/m^3) were observed in the western and eastern zones, with marked peaks in June and July up to 70% above average. This increase was related to the decrease in NO, which reduced O_3 titration and consumption, together with the occurrence of forest fires and anomalous solar radiation (Casallas et al. 2023).

In 2021, concentrations remained high in the peripheral areas and in the east of the city, with values above 30 µg/m^3 and 25 µg/m^3, respectively, showing a similar spatial distribution as in 2019. This may be attributed to the increase in NO and NO_2 emissions associated with economic

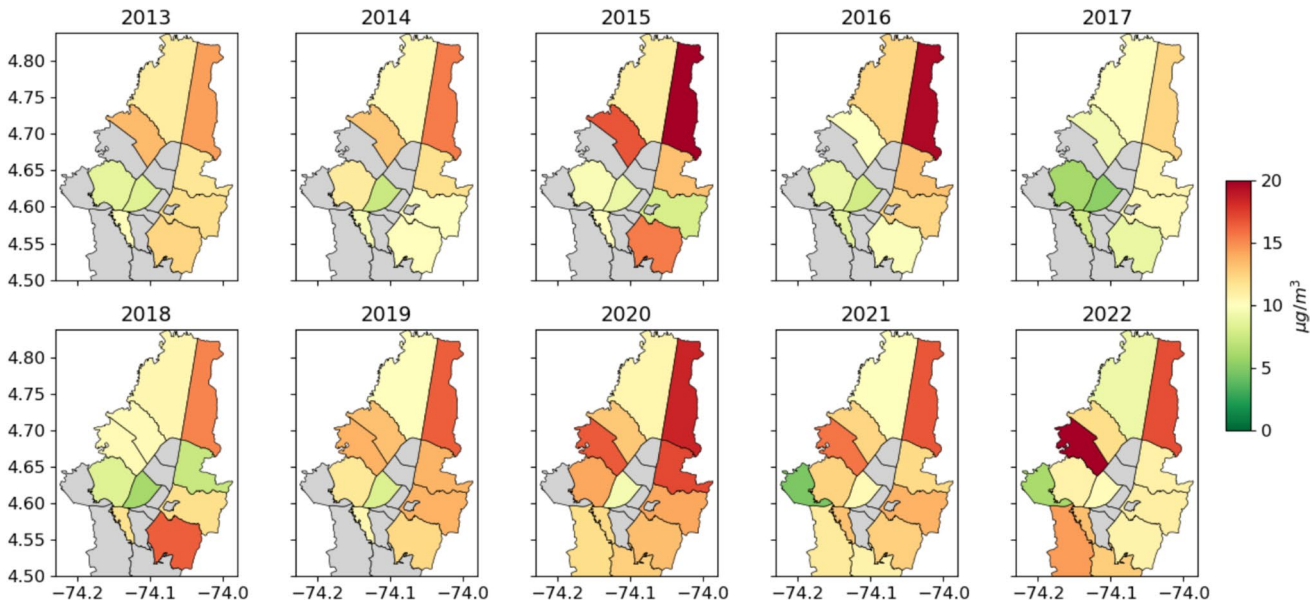


Fig. 2 Annual average of OSMDA8 in Bogotá, calculated from daily data of the maximum average of 8 hours in a day between 2013 to 2022. Grey polygons depict locations with less than 70% of available data for that year

reactivation after COVID restrictions. In 2022, a significant decrease in O_3 was observed in areas such as Kennedy (30 µg/m^3) and Fontibón (>25 µg/m^3), while in the east the values fell below 25 µg/m^3, but increased in Engativá (Figure SM1). This contrast may be attributed to local changes in industrial NO_x emissions, although more research is needed to fully understand this behavior.

It is also important to note that ozone formation tends to diminish in environments characterized by low solar radiation, high relative humidity, and weak winds, primarily due to limited regional O_3 advection. In Bogotá, low wind speeds combined with frequent cloud cover significantly reduce incoming solar radiation (Casallas et al. 2023), which may contribute to the observed suppression of ozone production, although further investigation is needed to confirm this relationship. Additionally, NO_x titration likely plays a role in the low ozone levels frequently recorded in the city (Sokhi et al. 2022). A negative correlation between ozone and other pollutants, such as particulate matter and CO, further supports the notion that these compounds do not substantially contribute to positive ozone formation in Bogotá's urban environment.

Importantly, while current ozone concentrations in Bogotá do not exceed the national regulatory threshold of 100 µg/m^3 for the OSMDA8 average (Resolution 2254 of 2017), and annual averages remain well below this level (typically not exceeding 45 µg/m^3) this should not be interpreted as an indicator of safety or stability. In fact, Colombia lacks a defined annual ozone standard, which limits the ability to assess long-term exposure risks. Comparatively, the WHO (2021) air quality guidelines recommend a much more stringent seasonal limit of 60 µg/m^3 (maximum 8-hour mean), suggesting that even moderate and increasing concentrations may pose a public health risk.

3.2 Seasonal Analysis

Figure 3 shows the monthly multiannual behavior of O_3 levels at different locations of Bogotá, calculated from the OSMDA8. The solid lines represent the monthly means, while the shaded areas show one standard deviation larger (smaller) than the mean, reflecting the interannual variability. A clear seasonal pattern can be seen at most stations, including CDAR, Usme, Tunal, Fontibón, Kennedy, Las Ferias, Usaquén, San Cristóbal, Puente Aranda and Minambiente, with ozone peaking between January and March (averages between 35 and 45 µg/m^3), followed by a decline between April and July (lows between 15 and 25 µg/m^3). From August, the levels rise again, reaching a second peak in September and October (30 to 40 µg/m^3), before falling in November and rising again in December.

Of all the stations, Usaquén has the highest ozone levels during seasonal peaks, exceeding 45 µg/m^3 in some months, while San Cristóbal has the lowest, with lows of around 15 µg/m^3 in the months with the lowest concentrations. This behavior suggests local differences influenced by topography, atmospheric ventilation, and sources emitting ozone precursors. The observed pattern shows the strong dependence of ozone on seasonal meteorological factors, such as solar radiation and wind dynamics, which condition its formation and dispersion (Cooper et al. 2014; Seinfeld And Pandis 2016; Casallas et al. 2024).

3.3 Multi-Year Ozone Trend Analysis

Figures 4 and SM2 presents the monthly temporal evolution of ozone concentrations in Bogotá from 2013 to 2023 (dependent on data availability), based on the OSMDA8. We separate the time series into its seasonal (purple line), trend (black line), and linear trend (red line) components (see methods for detail). A consistent seasonal cycle is observed across all monitoring stations, with ozone peaks typically occurring between January-March and September-November. These periods correspond to dry seasons in Bogotá, characterized by increased solar radiation and regional biomass burning, both of which enhance photochemical ozone production (Casallas et al. 2024).

Beyond this expected seasonality, the long-term trend component reveals significant spatial heterogeneity across the city, with the majority of the stations showing a trend increase (Fig. 4 and SM2). Several stations display clear and concerning upward trends. In contrast, some stations such as Carvajal (~ –0.05 µg/m^3/year), Jazmín (~ –0.65 µg/m^3/year), and Las Ferias (~ –0.22 µg/m^3/year) exhibit stable or slightly declining trends, likely due to the titration of ozone by high NO emissions from traffic and localized sources (Sokhi et al. 2022). Meanwhile, Guaymaral (~ 0.07 µg/m^3/year) and Centro de Alto Rendimiento (~0.03 µg/m^3/year) show relatively flat trends, suggesting either limited anthropogenic pressure or mitigating environmental factors like vegetation or enhanced ventilation.

These findings underscore two critical insights. First, the increasing trends observed in multiple urbanized stations (particularly those with vulnerable populations) highlight the emerging public health threat posed by ozone pollution in Bogotá. Second, the spatial variability in trends indicates that ozone cannot be addressed uniformly across the city; instead, tailored strategies are needed that consider local emission profiles and environmental conditions. The growing divergence between rising ozone levels in some areas and suppression in others suggests the city is undergoing a transitional phase in its air pollution dynamics.

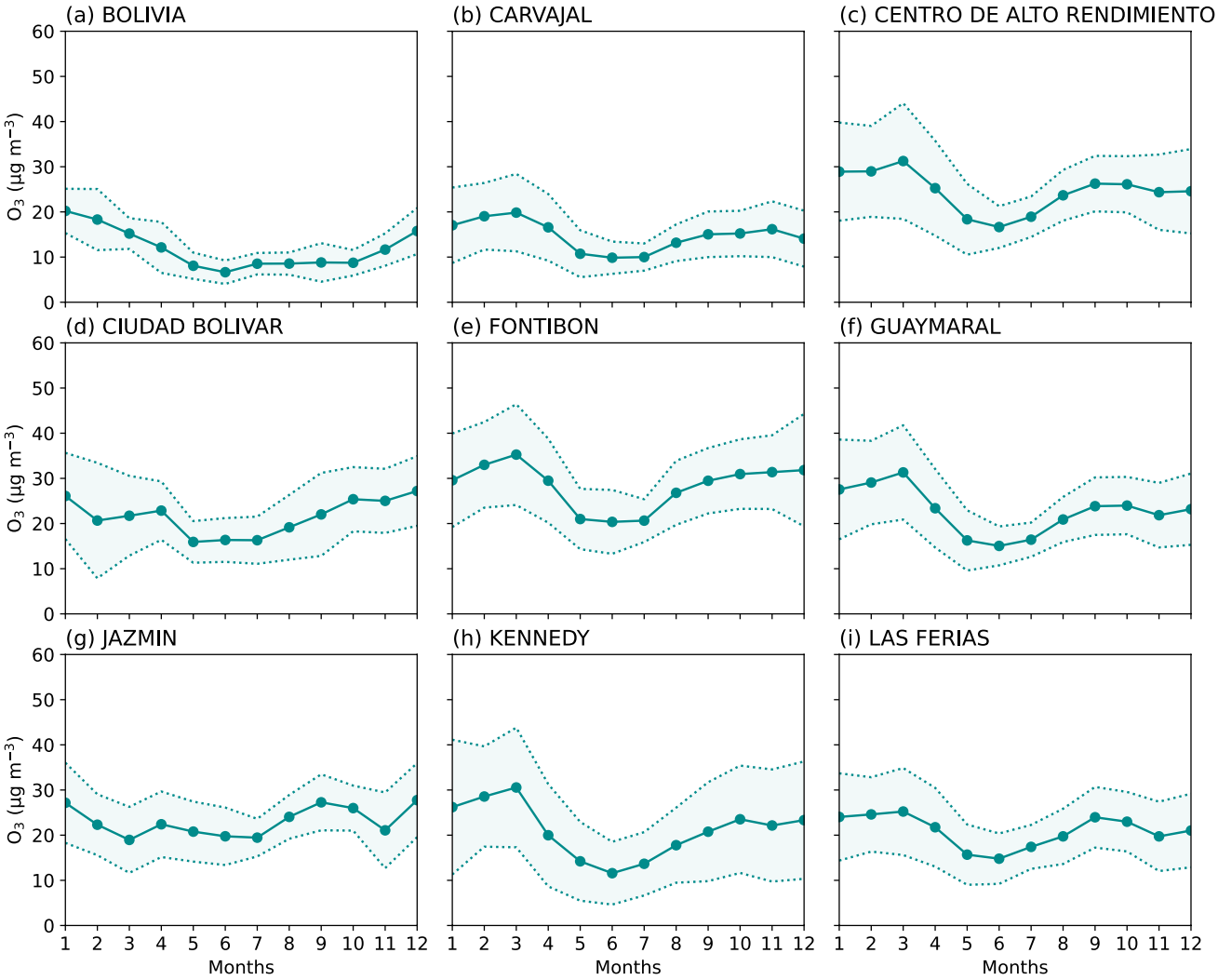


Fig. 3 Multi-year (2013–2023) monthly mean variations of tropospheric ozone levels calculated from the OSMDA8. Solid lines represent the mean and the shadow represents one standard deviation larger (smaller) than the mean

Our results indicate that ozone levels in most parts of the city have significantly increase over the past decade, signaling a troubling trend that may lead to broader urban-scale exceedances. This rise is particularly concerning given the well-established health risks of long-term ozone exposure (Murray et al. 2020). From a policy standpoint, sustained increases (especially in densely populated or disadvantaged areas) pose a growing environmental health burden (WHO 2021). These developments directly impact several Sustainable Development Goals (SDGs), including SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) (United Nations 2015). Integrating ozone monitoring and forecasting into local planning and establishing early warning systems (EWS) are critical steps toward achieving these goals (Lelieveld et al. 2015). Such measures enable proactive health protection, more resilient urban environments, and improved readiness for climate-related risks. Without them, policy responses remain reactive, limiting progress toward sustainable development.

Overall, these results reveal that ozone levels in Bogotá vary across the city and are influenced by a combination of local, meteorological, and emission factors. This emphasizes the importance of implementing context-specific monitoring and management strategies. Additionally, the results confirm an upward trend in ozone levels across the city, emphasizing the need for continuous monitoring and early warning systems to anticipate episodes of high concentrations, particularly in socio-environmentally vulnerable areas.

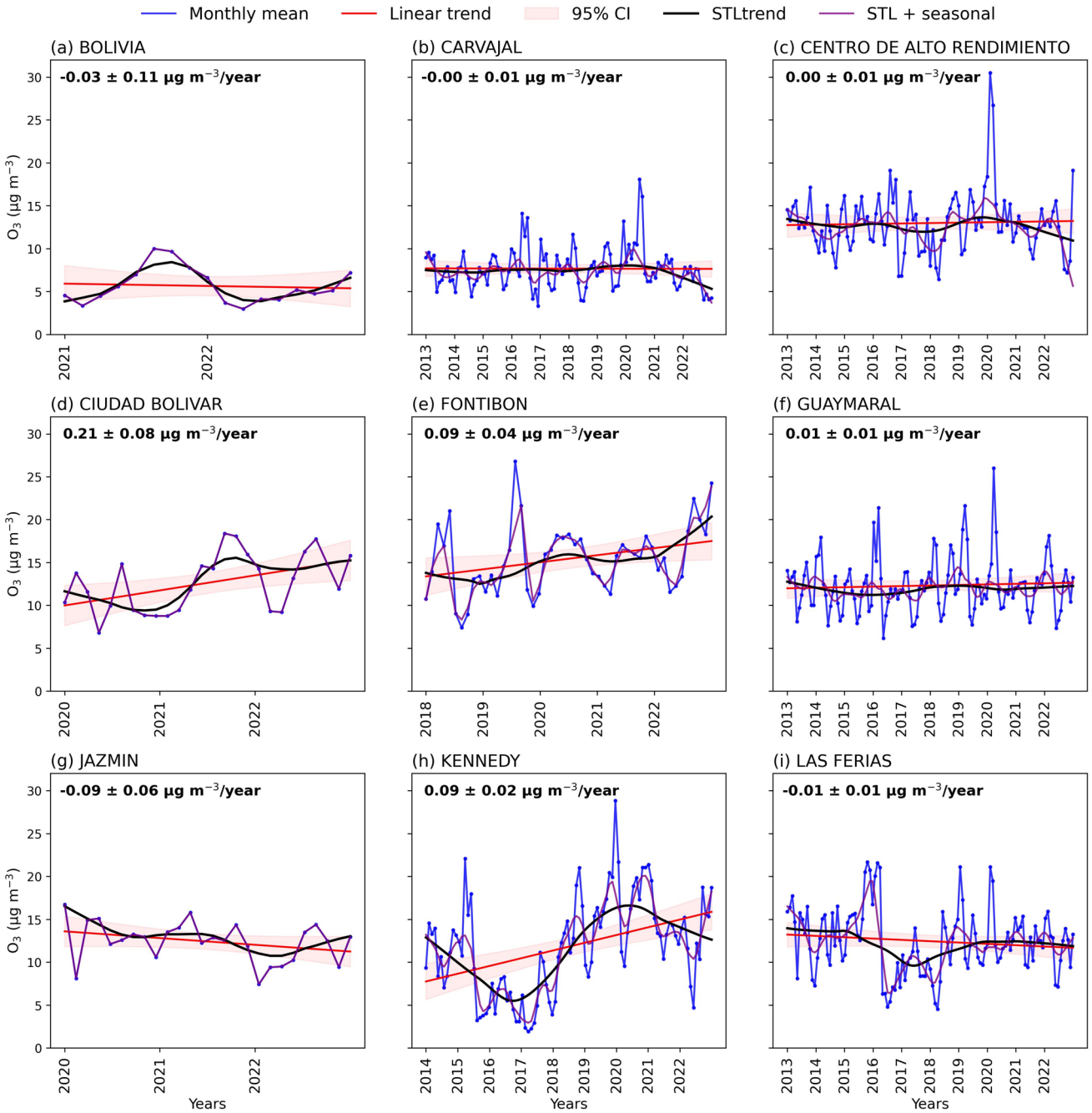


Fig. 4 Multi-annual trend of monthly ozone levels calculated from the OSMDA8 in Bogotá, obtained by STL decomposition and GLS regression. The purple line represents the seasonal component, the black line indicates the long-term trend, and the red line shows the GLS-adjusted slope reflecting the general linear trend of ozone over 2013-2023 (dependent on data availability), with the red shading depicting the 95% confidence interval (CI)

3.4 Public Health: Deaths Attributable to Ozone Exposure

From historical data from the District Secretary of Health (2013–2023), nine diseases associated with ozone mortality (Malashock et al. 2022) were studied (see methods). (i) COPD = 49.9%, (ii) ALRI= 26.89%, (iii) COPD-AE= 11.36%, (iv) CVD = 8.53%, (v) Other lung disease = 1.94%, (vi) IHD= 1.04%, (vii) acute lower respiratory infection= 0.3%, (viii) lung cancer mortality= 0.1%, and (ix) other infections= 0.02%) (Figure SM3).

Figures 5 and 6 shows the spatial and age distribution of ozone-attributable mortality in Bogotá (per 1000 inhabitants), with clear patterns in both age and geographic

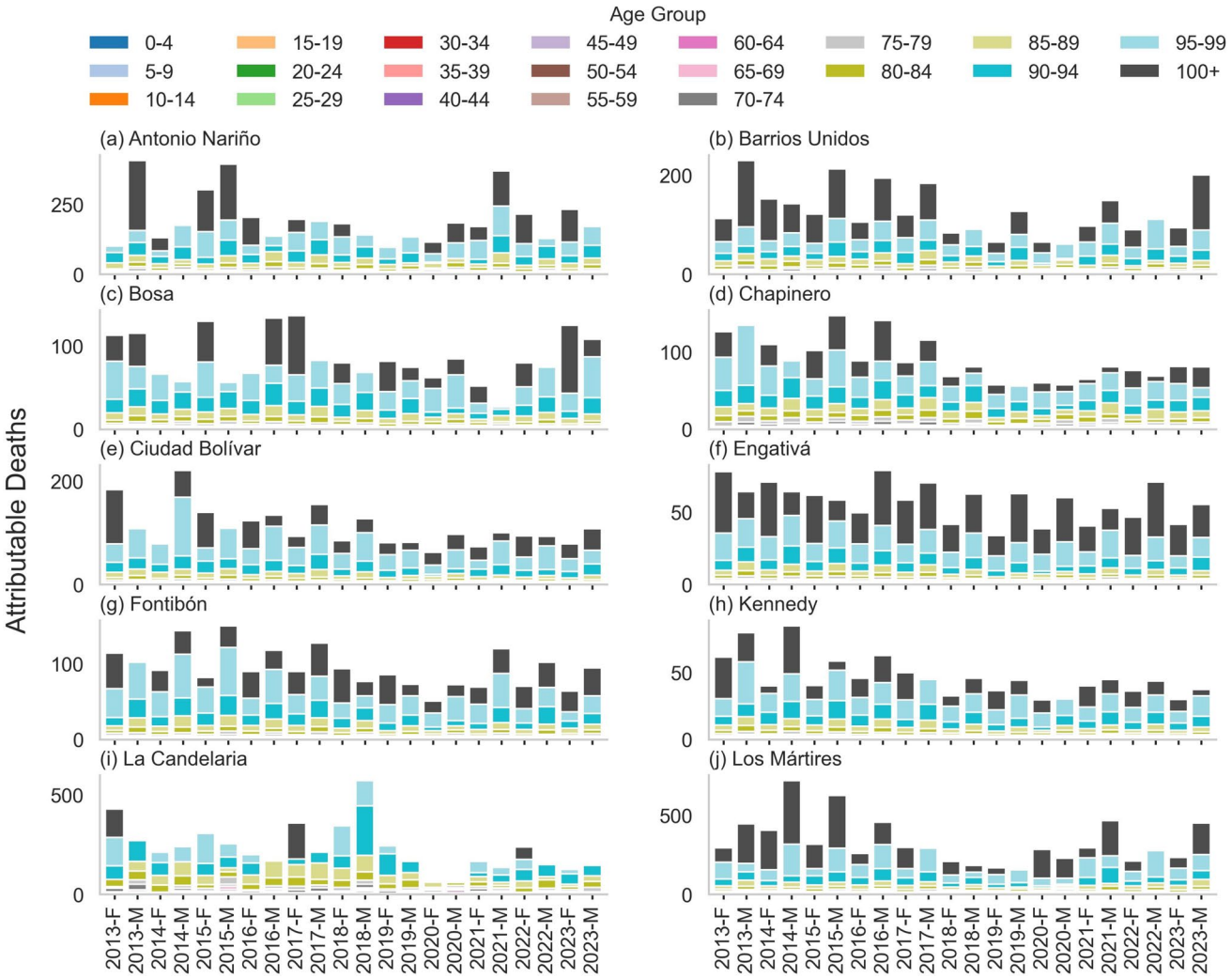


Fig. 5 Number of attributable deaths due to ozone exposure per 1000 inhabitants in 10 localities of Bogotá for the age group and sex between 2013 and 2023. The letter adjacent to the year in the x-axis represents the sex, being F for females and M for males

location. The data show that the risk of dying from ozone exposure increases significantly with age, with the 65+ age group being the most affected in almost all localities (Bonilla et al. 2023; Pozzer et al. 2023). In localities such as Suba, Kennedy, and Engativá (see Fig. 1 for the geographic location) this group exceeds 100 deaths per year (per 1000 inhabitants), peaking at up to 200 deaths in Kennedy (Figs. 5 and 6).

During the period 2013–2023, the mortality burden attributable to ozone was relatively low in the younger age groups (children, 0–14 years, and young adults, 15–44 years) in Bogotá (Figs. 5 and 6). In most areas, the number of deaths in these age groups remained below 10 per year, even in areas with high exposure, such as Kennedy, Suba, and Ciudad Bolívar. This low mortality can be attributed to the lower prevalence of chronic diseases and greater physiological response to exposure in these age groups. However, it should be noted that these indicators do not necessarily reflect subclinical effects or long-term impacts, such as the progressive deterioration of lung function, increased systemic inflammation, or greater susceptibility to disease in adulthood (Chen et al. 2023). This is particularly pertinent in the case of children, whose respiratory systems are still developing.

In contrast, the 45–64 age group shows an increasing trend in ozone-attributable mortality over the analyzed period, indicating an emerging risk group. In localities such as Ciudad Bolívar, Kennedy, and Suba, 30–50 deaths per year (per 1000 inhabitants) are recorded from this cause (Figs. 5 and 6), indicating a sustained increase in this age group as prolonged exposure accumulates and chronic conditions such as cardiovascular disease, chronic obstructive pulmonary disease (COPD), or lung cancer manifest. While the mortality burden in this group does not reach the levels observed in those over 65+ age group (Figs. 5 and 6), it is significant from a public health perspective as this group

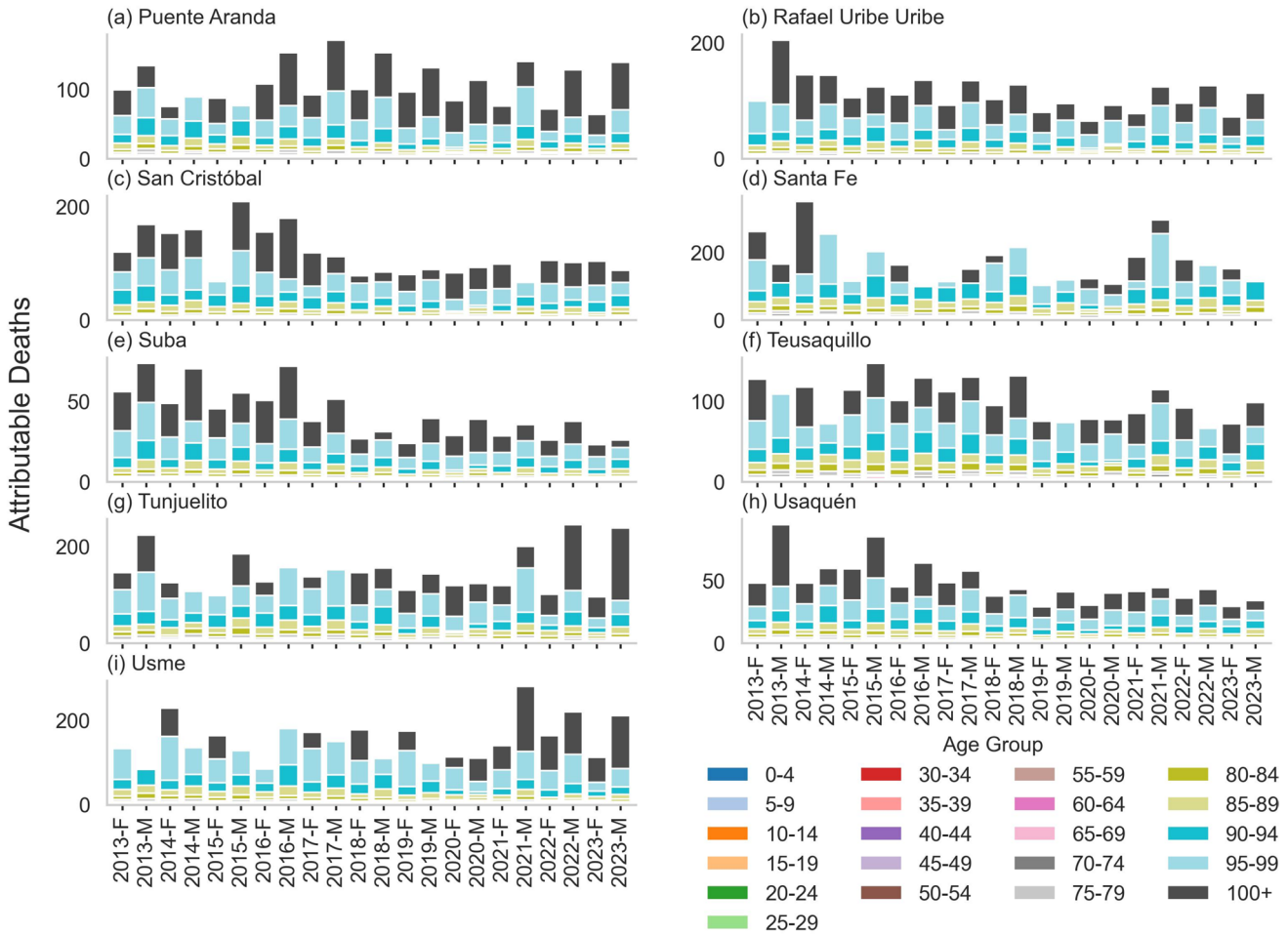


Fig. 6 Number of attributable deaths due to ozone exposure per 1000 inhabitants in 9 localities of Bogotá for the age group and sex between 2013 and 2023. The letter adjacent to the year in the x-axis represents the sex, being F for females and M for males

is economically active and productive. These findings reinforce the need for early preventive interventions and the implementation of monitoring and warning systems that focus not only on older adults but also on this age group, which is beginning to show clinically relevant effects of ozone exposure.

Spatially, the areas most affected by ozone-related mortality are Suba, Kennedy, Engativá, and Ciudad Bolívar, which combine high ozone concentrations with high population density and possibly less access to health services. In contrast, areas such as La Candelaria, Santa Fe, and Teusaquillo have the lowest rates, with at least less than five deaths per year in all age groups. These results highlight the spatial inequality of ozone-related health risks in Bogotá, with older adults and densely populated peripheral areas more affected, and call for targeted public health and air quality interventions (Figs. 5 and 6). The data show that the ozone mortality risk increases progressively between 2013 and 2019, especially in older age groups (≥65 years) and in peripheral areas with high population density.

From 2020 to 2023, a slight decrease in ozone-attributable mortality is observed in some areas. This may be related to reduced population exposure resulting from temporary declines in traffic and industrial activity during the COVID-19 pandemic, rather than an actual reduction in ozone concentrations (Figs. 5 and 6). However, in certain localities (e.g., Tunjuelito, Bosa, Barrios Unidos, Usme, and Puente Aranda) an increase in mortality began to emerge during 2022 and 2023. This suggests that, although ozone levels continued to rise, the temporary reduction in population exposure during the pandemic led to a short-term decline in ozone-attributable mortality. However, as mobility and economic activities resumed, several localities began to show a renewed increase in mortality during 2022 and 2023, indicating a rebound effect, probably linked to the return of exposure rather than changes in ambient ozone concentrations.

In summary, the 2013-2023 data show the high risk of older adults and also a clear spatial inequality, with higher risk in peripheral areas in the south and west of Bogotá. This

pattern highlights the need for environmental health policies that target both age and territory, including measures to control emissions and strengthen health care for the most vulnerable groups.

3.5 Health Impacts from Prolonged Ozone Exposure Include Mortality from Diseases

The RR of mortality associated with ozone exposure was estimated using the hazard ratio (HR) derived from the comprehensive GEMM, as illustrated in Fig. 7. The HR values are presented alongside their 95% confidence intervals (CIs), providing a measure of statistical reliability for the risk estimates.

The findings indicate that relative mortality risk increases progressively with higher ozone concentrations (Fig. 7). This trend suggests a direct and statistically significant association between ambient O_3 levels and mortality risk, particularly when the confidence intervals do not include one. These results underscore the health implications of rising ozone exposure and the importance of ongoing monitoring and mitigation efforts.

The application of the GEMM model strengthens the validity of these results, as it allows the inclusion of a large global epidemiological database and the adjustment for multiple confounding variables. Taken together, the data presented in Fig. 8 support the hypothesis that long-term exposure to ozone is an important risk factor for mortality and demonstrate the need for more stringent environmental control measures. The data provides key evidence of the differential health impact of tropospheric ozone in Bogotá (Fig. 8). It shows the RR values for various specific causes of death that are associated with prolonged exposure to this pollutant. The results reveal substantial variability in the magnitude of the risk depending on the physiological system affected, showing that the effect is far from homogeneous. This has direct implications for prevention and health alert strategies.

The strongest associations are particularly evident in chronic respiratory and cardiovascular diseases, which is consistent with international literature (e.g., Bonilla et al. 2023; Pozzer et al. 2023). The highest relative risk (RR = 1.36; 95% CI: 1.28–1.44) is presented by COPD, followed by IHD with an RR of 1.34 (95% CI: 1.30–1.38) (Fig. 8). These values indicate that prolonged exposure to ozone increases the risk of death from these conditions by over 30% compared to exposure at lower levels. This is consistent with the findings of Bell et al. (2006), who concluded that there is no safe threshold for ozone, as its effects on mortality can be observed at concentrations below regulatory limits. Other conditions, such as lung cancer (RR = 1.18), lower respiratory infections (RR = 1.20), and cerebrovascular accidents (RR = 1.22) (Fig. 8), also demonstrate statistically significant, albeit lesser, increases in risk. These findings demonstrate that, while ozone primarily affects the respiratory system, it also exacerbates cardiovascular and systemic diseases (Malashock et al. 2022).

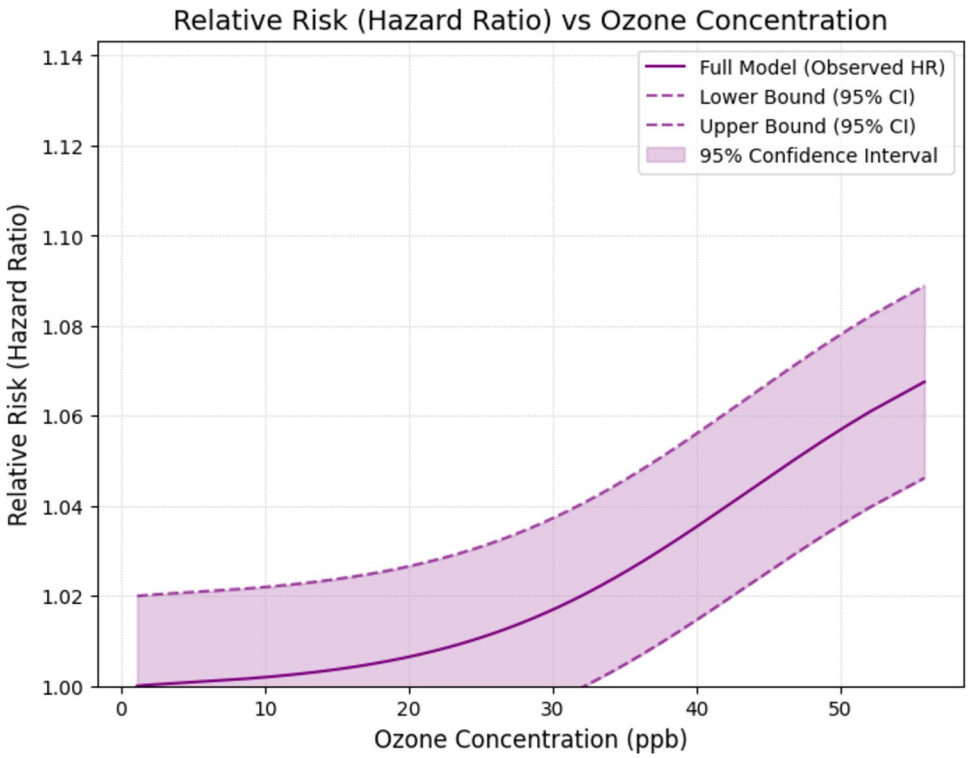


Fig. 7 Relative risk with an HR. An interval of 95% of confidence intervals was calculated with GEMM

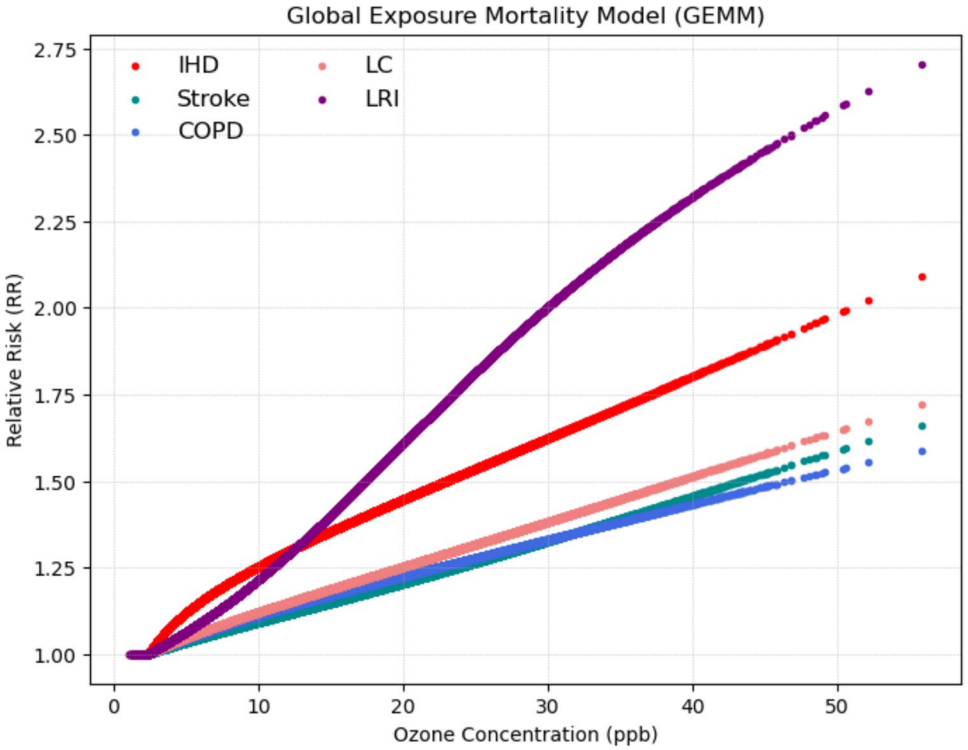


Fig. 8 Relative risk of ozone deaths in Bogotá for five diseases calculated with GEMM. IHD: Ischemic heart disease, COPD: Chronic Obstructive Pulmonary Disease, LC: Lung cancer, LRI: lower respiratory illness diseases, and Stroke: namely stroke. An interval of 95% of confidence intervals was calculated with GEMM

This pattern of differential risk reinforces the need for targeted, proactive mitigation strategies. EWSs should prioritize the most vulnerable populations, particularly older adults and individuals with pre-existing health conditions. Public policies must better integrate health and air quality considerations to ensure effective protection. Furthermore, the data shown in Fig. 8 underscores the importance of strengthening continuous ozone monitoring and highlights the critical need to measure VOCs, which play a key role in ozone formation. It also points to the necessity of establishing stricter regulatory limits that account not only for concentration levels but also for exposure duration and cumulative population vulnerability (Celis et al. 2022).

Building on the RR estimates shown in Fig. 7 and 8, Fig. 9 presents the annual evolution and spatial distribution of mortality attributable to ozone exposure in Bogotá, disaggregated by sex, locality, and year. The results reveal a variable but generally increasing trend in ozone-attributable deaths over the period 2013-2023, with notable increases in certain localities and specific years. The highest burden appears between 2010 and 2019, while a slight decrease is observed in 2020 and 2021, likely reflecting the impact of reduced human activity and mobility restrictions during the COVID-19 pandemic. However, in 2022 and 2023, a renewed increase in ozone-attributable mortality is observed in several localities (e.g., Tunjuelito, Bosa, Barrios Unidos, Usme, and Puente Aranda) coinciding with the resumption of traffic, economic activity, and thus greater population exposure.

Ozone-attributable mortality in Bogotá showed an upward trend from 2013 to 2019, followed by a slight decline in 2020–2021 and then a renewed increase in 2022 and 2023 as normal activities and associated emissions resumed after the lockdowns. The highest mortality burdens are concentrated in localities such as La Candelaria, Los Martires, Ciudad Bolívar, and Barrios Unidos, areas that may reflect heightened social vulnerability, cumulative exposure, substandard living conditions, and limited access to healthcare. Overall, men and women are similarly affected across most localities, with no statistically significant differences between sexes (Fig. 9). These findings underscore the need for environmental health policies that integrate spatial and socioeconomic perspectives to address disparities in the impacts of ozone pollution.

To quantify the health impact of prolonged ozone exposure, we use the total load, which is the long-term average OSMDA8 ozone concentration, calculated over the period 2013-2023. This metric is then applied using the method proposed by Burnett et al. (2018) to estimate the proportion of deaths attributable to long-term ozone exposure among individuals aged 25 and older, disaggregated by specific causes of death. The resulting estimates, presented in Table 2, provide a clearer assessment of the disease burden associated with chronic ozone exposure across various demographic groups.

These estimates indicate that about 18.3% of deaths from all causes in the population of Bogotá could be related to ozone exposure. A significant proportion of deaths from

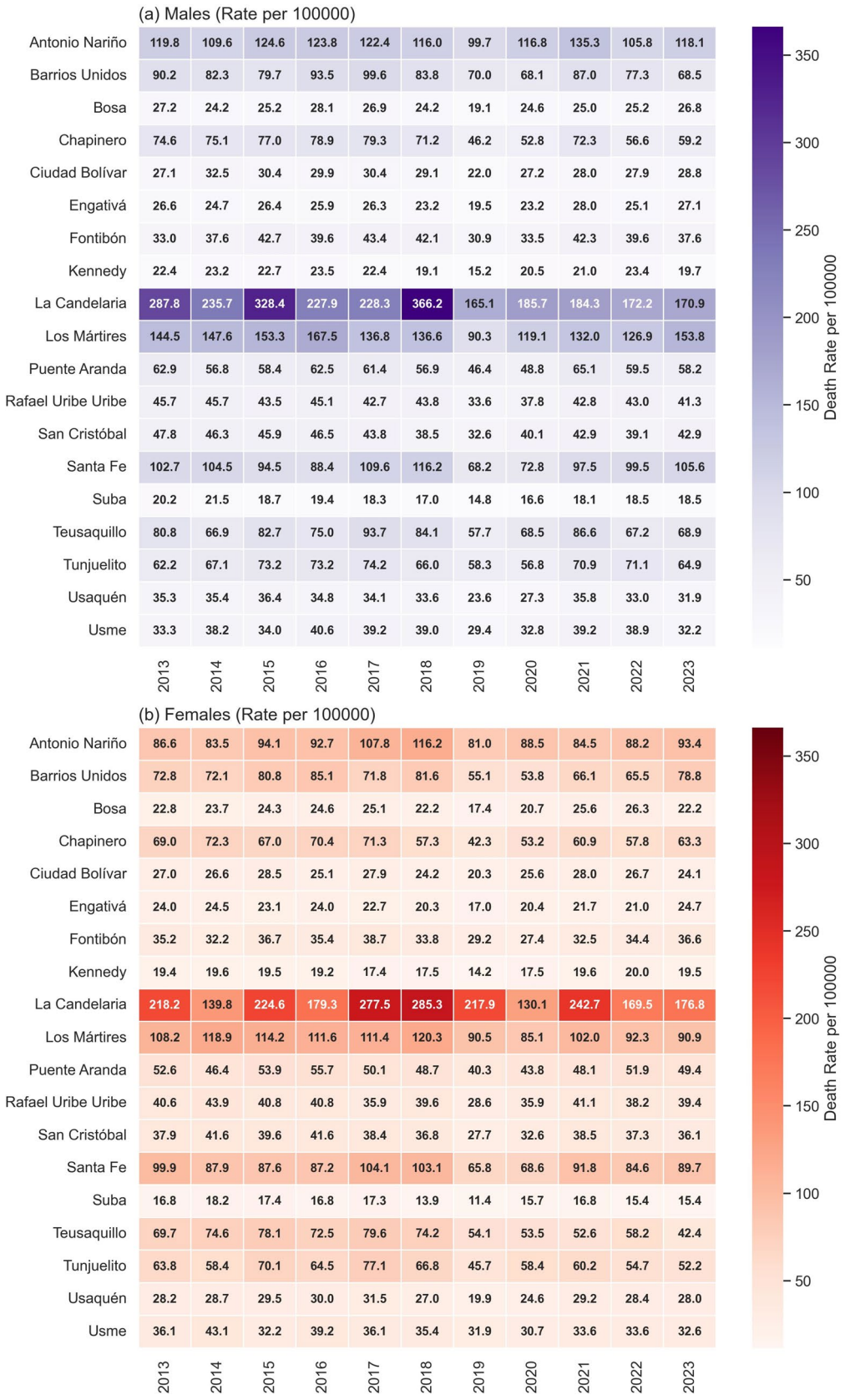


Fig. 9 Number of deaths attributable to ozone exposure for every 100000 inhabitants in all the localities of Bogotá for (**a**) males and (**b**) females over 2013-2023

Table 2 Fraction of deaths attributable to ozone exposure estimated according to the methodology of Burnett et al. (2018) for Bogotá

Cause of death	Attributable fraction (median)
All-cause mortality	17.6%
Total respiratory mortality	1.38%
COPD mortality	1.11%
Cardiovascular mortality (CVD)	0.12%
Ischaemic heart disease (IHD) mortality	2.39%
Non-communicable disease (NCD) mortality	6.23%

noncommunicable diseases (6.23%) and ischemic heart disease (2.39%) could also be attributed to this pollutant. Although the attributable fractions for specific causes such as COPD and total cardiovascular mortality are lower, they still represent a significant public health burden (Table 2). These results are consistent with the estimates from the WHO's Global Burden of Disease (GBD) project, which reports that long-term ozone exposure accounts for approximately 0.5-1.0% of all global deaths and contributes to around 400000 premature deaths annually worldwide (Murray et al. 2020; WHO 2023). The similarity between our local attributable fractions and these global estimates supports the robustness of our findings and indicates that Bogotá's ozone-related mortality burden aligns with international patterns when adjusted for population size and exposure levels. Likewise, we present the estimated absolute number of deaths attributable (per 100000 inhabitants) to air pollution in the city of Bogotá and its localities, calculated from exposure-response functions, for people aged 25 years and older. The 95% CIs are also included (Fig. 10).

Results show the spatial inequalities in the mortality burden attributable to ozone in Bogotá, adjusted for population size. Expressing the estimates as deaths per 100000 inhabitants over the age of 25 reveals that the southern districts have the highest rates, exceeding 14 deaths per 100000 inhabitants in total attributable deaths (Fig. 10). Kennedy, Ciudad Bolívar, Puente Aranda, and Teusaquillo, which are located in the Bogotá downtown and southwestern areas (see Fig. 1 for the geographic location of the stations), have the highest combined rate, coinciding with the findings of Bonilla et al. (2023). This suggests a combination of high exposure and a particularly vulnerable population.

This pattern indicates an uneven distribution of ozone impacts, with the area's most vulnerable coinciding with those experiencing the highest attributable mortality (Bonilla et al. 2023). This highlights the need for territory-focused environmental mitigation and public health prevention strategies in the southern and south-western areas of the city, with special attention given to these regions. The associated health burden is the result of a combination of environmental and socioeconomic factors. These areas have a high population density, implying a higher potential exposure. They also have intense commercial, industrial, and vehicular traffic activity, favoring the emission of ozone precursors such as NO_x and VOCs (Seguel et al. 2024). These localities also have social conditions that can exacerbate the effects of ozone, such as limited access to healthcare and a more vulnerable population (Ramirez et al. 2019; Malashock et al. 2022).

In contrast, Tunjuelito, Usme, and Usaquén show the lowest number of attributable deaths (Fig. 10), potentially due to factors such as greener space, or lower baseline ozone concentrations (DANE 2025). However, even in these relatively "cleaner" areas, mortality attributable to ozone is still present, indicating that the health effects of ozone are widespread, though variable in magnitude. From

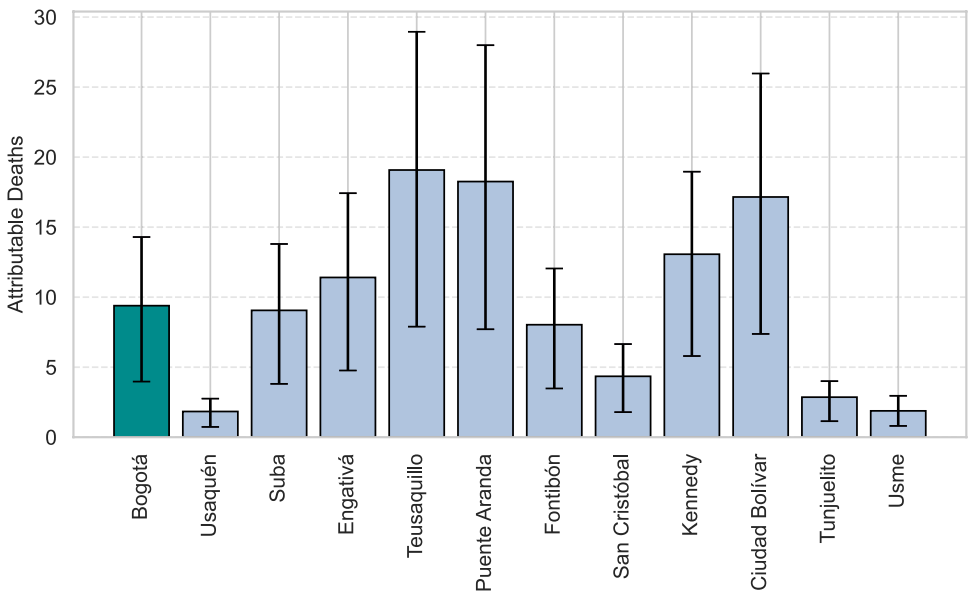


Fig. 10 Number of deaths attributable to ozone exposure per 100000 inhabitants in Bogotá (dark cyan bar), by locality (light grey bars), for individuals aged 25 and older. The bars represent the 95% confidence interval for each estimate

a territorial perspective, the southwestern localities, especially Kennedy, Ciudad Bolívar, and Teusaquillo, emerge as critical areas for intervention due to their combined environmental and demographic vulnerabilities. These findings suggest that efforts to improve air quality should be geographically targeted, prioritizing communities with the highest exposure and burden.

To address these disparities, we recommend that public health authorities in Bogotá prioritize interventions in the most affected districts. These may include implementing stricter emissions controls on local transport and industry, expanding green infrastructure to absorb pollutants and if possible, also mitigate heat (Moreno-Cortés et al. 2021), and improving access to healthcare and early-warning systems in vulnerable communities (Celis et al. 2022; Casallas et al. 2022). In addition, targeted air quality monitoring and public awareness campaigns can enhance responsiveness and resilience in these high-risk zones.

Overall, our analysis shows that approximately 18.3% of all deaths among individuals aged 25 and older in Bogotá can be attributed to long-term exposure to ozone and other air pollutants. This proportion aligns with global evidence indicating substantial mortality burdens linked to ozone, even at concentrations below current regulatory thresholds (Pozzer et al. 2023). For instance, Bell et al. (2006) found that there is no safe threshold below which ozone has no effect on mortality; even a modest 10 ppb increase in ozone was associated with a 0.30% rise in daily mortality. These findings are reflected in our data, where elevated mortality is observed even in areas where ozone levels often remain within national standards.

Similarly, Seguel et al. (2024) reported a gradual increase in long-term ozone exposure in Bogotá, particularly in the northern and central zones. Our spatial analysis complements this observation by showing that localities in the southwest and northwest, while more industrialized and densely populated, are currently the most affected by ozone-related mortality, suggesting a convergence between increasing background exposure and local risk factors. At the regional scale, Malashock et al. (2022) documented a 46% global increase in ozone-attributable deaths between 2000 and 2019, with Latin America experiencing a disproportionate rise of 350%. Our findings place Bogotá squarely within this trend: as a rapidly growing urban center, it faces intensifying environmental health challenges driven by population growth, increased motorization, and the concentration of emissions in vulnerable neighborhoods (Bonilla et al. 2023). Moreover, ozone levels have been increasing across nearly the entire city (Fig. 4), further exacerbating the public health burden and illustrating the importance of city-wide and locally targeted interventions.

When compared with international benchmarks such as the WHO's Global Burden of Disease estimates, the ozone-attributable mortality fraction in Bogotá is within the expected range for highly urbanized regions of Latin America, where values between 10% and 20% of all-cause mortality have been reported due to ambient air pollution (Murray et al. 2020; WHO 2023). This further reinforces the external validity of our results and underscores the need for locally tailored interventions consistent with global health risk assessments. Taken together, this body of evidence highlights the urgency of locally adapted policy responses. In Bogotá, mitigation strategies should focus on high-burden localities, incorporating lessons from cities like São Paulo and Santiago, where sustained interventions have helped stabilize ozone levels (Seguel et al. 2024). Importantly, our findings stress that even moderate levels of ozone exposure pose a real and measurable risk to public health, particularly in the most populated and polluted areas of the city.

Nonetheless, it is important to acknowledge that residual confounding may still influence the estimated ozone–mortality associations. While our short-term analysis using the ERF model included adjustment for major time-varying confounders such as seasonality, meteorology, and long-term trends (following Bell et al. 2005), and our long-term burden estimates relied on hazard ratios derived from large cohort studies that account for key individual-level factors such as age, sex, smoking, and socioeconomic status (Burnett et al. 2018), the lack of direct access to individual-level data in our mortality dataset limits our ability to fully control for all potential sources of bias. However, both Bell et al. (2005) and Burnett et al. (2018) conducted extensive sensitivity analyses demonstrating that the estimated ozone–mortality relationships remain robust under alternative model specifications, confounder adjustments, and lag structures. This provides additional confidence in the methodological soundness of our approach. Nevertheless, this limitation is inherent to ecological studies and should be considered when interpreting the results, particularly regarding the magnitude of the estimated effects.

3.6 Evaluation of the Neural Network Model

Given the increasing trends in ozone concentrations and the associated health burden identified across Bogotá, it is critical to develop tools that can anticipate hazardous air quality conditions before they occur. In response to this need, we implemented a neural network (NN, see method for details), with the goal of predicting hourly ozone levels as a basis for early warning systems.

Figure 11 illustrates the temporal agreement between predicted and observed ozone concentrations across six representative stations during a 100-hour period (last 100 days

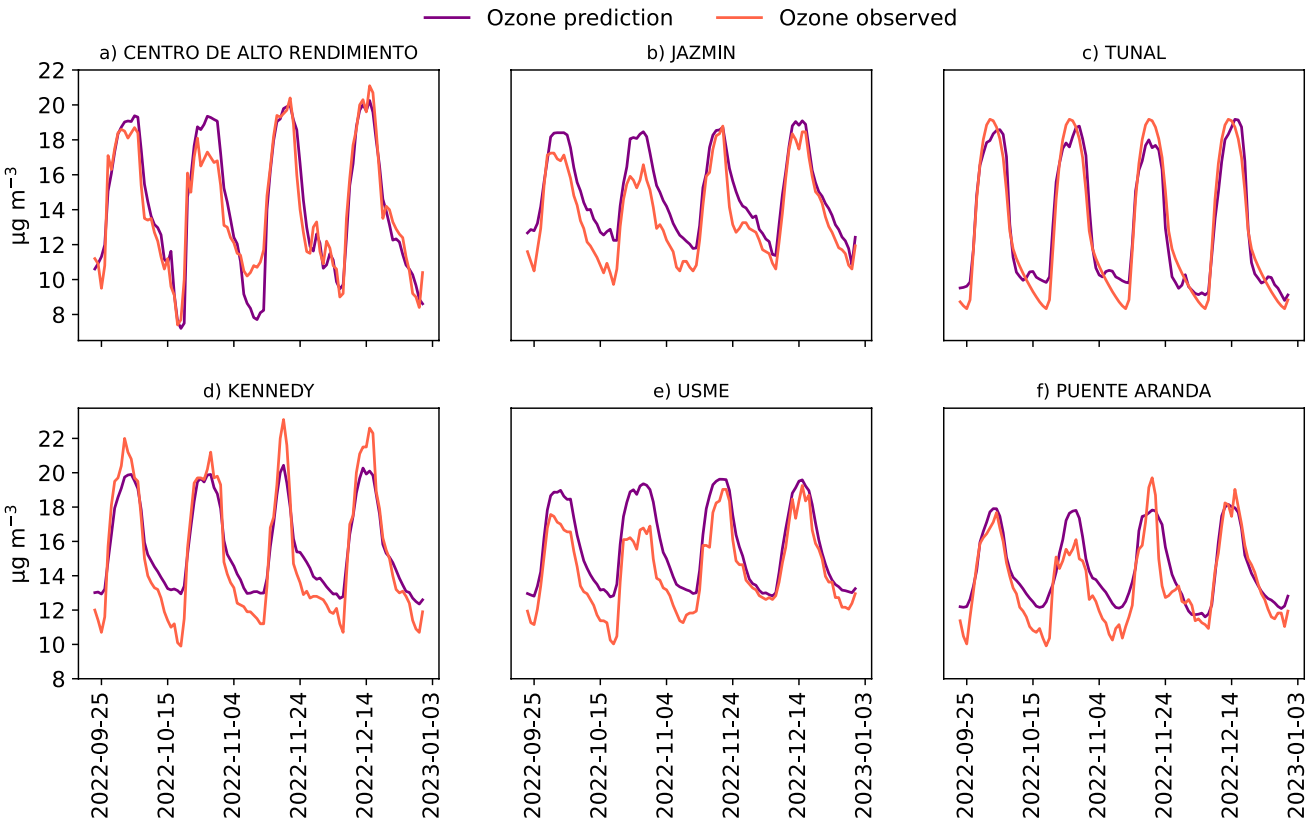


Fig. 11 Comparison between observed and predicted ozone concentration values at the selected RMCAB stations. The purple line is the predicted ozone and the red line is the observed ozone

of December 2022). The model captures the diurnal pattern of ozone evolution well, reproducing the timing and relative amplitude of peaks. However, some underestimation of peak values is evident, particularly in high-emission zones such as Kennedy and Puente Aranda. The NN model used here appears to offer a more conservative estimate under rapidly changing conditions but struggles with sharp peaks, which may result from episodic emissions or abrupt meteorological changes not fully captured by the input data (Mogollón-Sotelo et al. 2021). This limitation could be addressed by designing a specialized loss function that penalizes underprediction of peak events more heavily, thereby improving the model's sensitivity to extreme values (Cache et al. 2024; Rampal et al. 2024). Additionally, incorporating stochastic components (e.g., probabilistic forecasting or ensemble approaches) may help the model better represent uncertainty and variability in ozone formation during atypical conditions, such as pollution spikes or the presence of plumes from wildfires (Grundner et al. 2024; Lin et al. 2025), although this is out of the scope of this paper.

Figure SM4 presents results from stations lacking a complete set of input variables. Despite this limitation, the model successfully captured the overall trend and behavior in most cases, particularly during the last days peaks, where differences were typically under 1 µg/m^3. At the Las Ferias, Min Ambiente, and San Cristóbal stations, the model overestimated concentrations during the first two peaks, with discrepancies exceeding 1 µg/m^3. However, it still reproduced the general temporal pattern observed across the stations, minimizing the overall deviation. For the Usaquén station, while initial values were also overestimated, the model achieved greater accuracy during the last peak, with a difference of only about 1 µg/m^3 (19.4 µg/m^3 observed vs. 20.5 µg/m^3 predicted). Ciudad Bolívar showed the lowest peak values in both predictions and observations, with a maximum of 17 µg/m^3 (Figure SM4). The stations that reached predicted values above 20 µg/m^3 were Carvajal, Suba, and Usaquén, successfully replicating the highest observed peaks. Similarly, the Suba station recorded predictions below 9 µg/m^3 between November 24 to 27, matching the observed values.

The scatter plots in Fig. 12 provide a station-by-station comparison of predicted versus observed ozone values, highlighting the model's performance across the full range of concentrations (the other stations can be seen in Figure SM5). Most points are tightly clustered along the 1:1 line, with strong linear correlations between observed and predicted values. Notably, some underestimation occurs

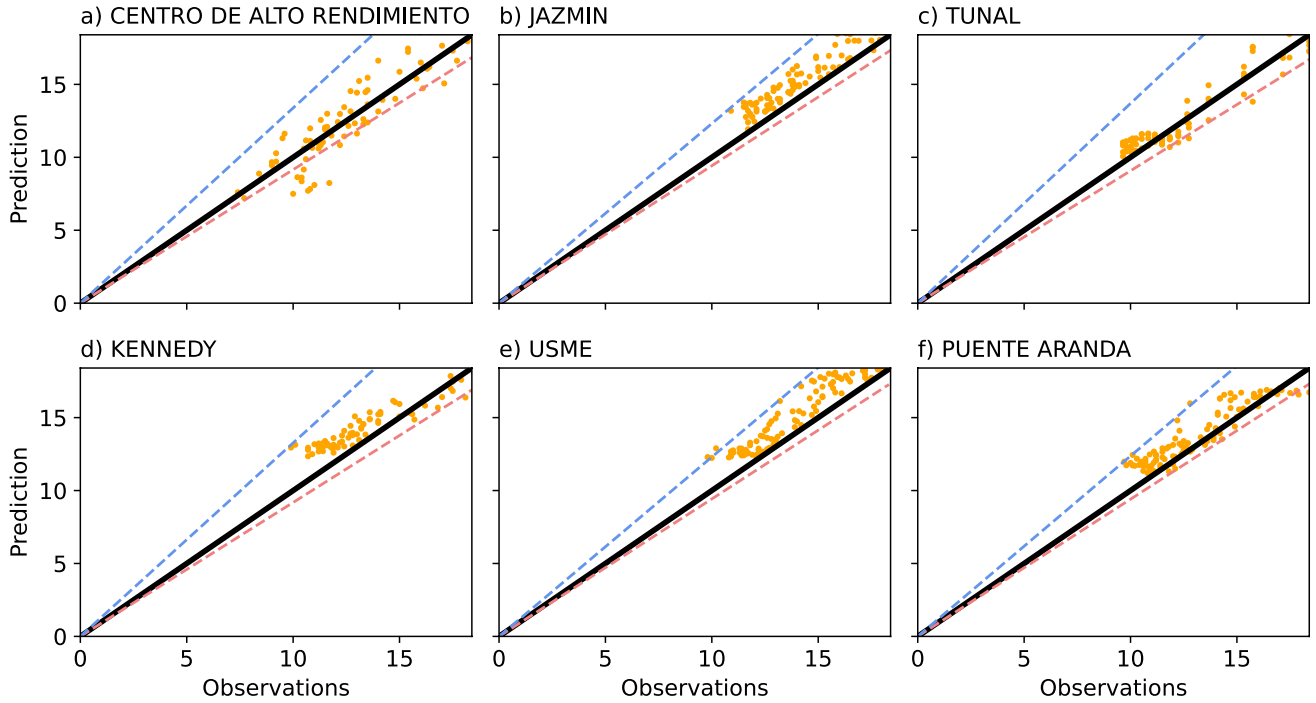


Fig. 12 Statistical validation of the ozone prediction model using the Rho correlation for ozone (units of $\mu g/m^3$) predicted in six municipalities. The black line corresponds to the 1:1 linear regression; the blue and red dotted lines correspond to 1:2 and 1:0.5, respectively

at moderate concentrations (10–15 $\mu g/m^3$), especially at Centro de Alto Rendimiento and Puente Aranda. These discrepancies may stem from the model's difficulty in resolving local emission spikes or a lack of precursors such as NMHCs, which were not included due to data unavailability but play a critical role in ozone formation (Hanif et al. 2021; Eslhorbany et al. 2024).

To robustly evaluate the model's performance, we conducted a full statistical validation summarized in Fig. 13 (see also Figure SM6). Across most stations, the model performs well, with Rho values above 0.6, indicating strong agreement between predictions and observations. The highest Rho is observed at Guaymaral (0.87), reflecting the model's ability to track ozone in less polluted environments. Conversely, stations with denser urban activity and more variability, such as Puente Aranda and Kennedy, show lower scores, emphasizing the challenge of modeling ozone in heterogeneous emission environments. From a categorical perspective, the model achieves POC values above 60% and HIT rates above 50% in many cases, suggesting strong potential for early detection of elevated ozone levels. The FAR remains high at several stations, indicating a tendency to overpredict exceedance events, an acceptable compromise in the context of public health, where missed warnings are more critical than false alarms.

Collectively, these results confirm that the NN model is capable of capturing temporal dynamics and spatial variability in ozone concentrations across Bogotá. While some limitations remain, particularly in resolving extreme values, the model provides a promising foundation for operational EWSs. Further improvements could be achieved by increasing the number of training epochs and incorporating additional predictors such as NMHCs, traffic volume, or real-time satellite data (Abdul-Wahab And Al-Alawi 2002).

These validation results support the integration of the CNN-BiLSTM model into an operational ozone EWS for Bogotá. By using real-time inputs from the existing air quality monitoring network and meteorological stations, the trained model can generate short-term forecasts of ozone concentrations across the city. As demonstrated in our evaluation, even in complex urban environments, the model performs with sufficient accuracy to flag periods of elevated ozone risk, providing actionable lead time for decision-makers. Within this operational context, data-driven approaches offer a practical complement to more traditional modeling frameworks.

Regional chemical transport models remain essential for process-based understanding of ozone formation and transport, but machine learning methods such as the CNN–BiLSTM are particularly well suited for station-level prediction and rapid forecasting applications. Because these approaches rely on fundamentally different assumptions, spatial representations, and input data, direct comparisons with chemical transport models should be interpreted with caution. Nevertheless, the results presented here highlight the potential of machine learning methods to efficiently

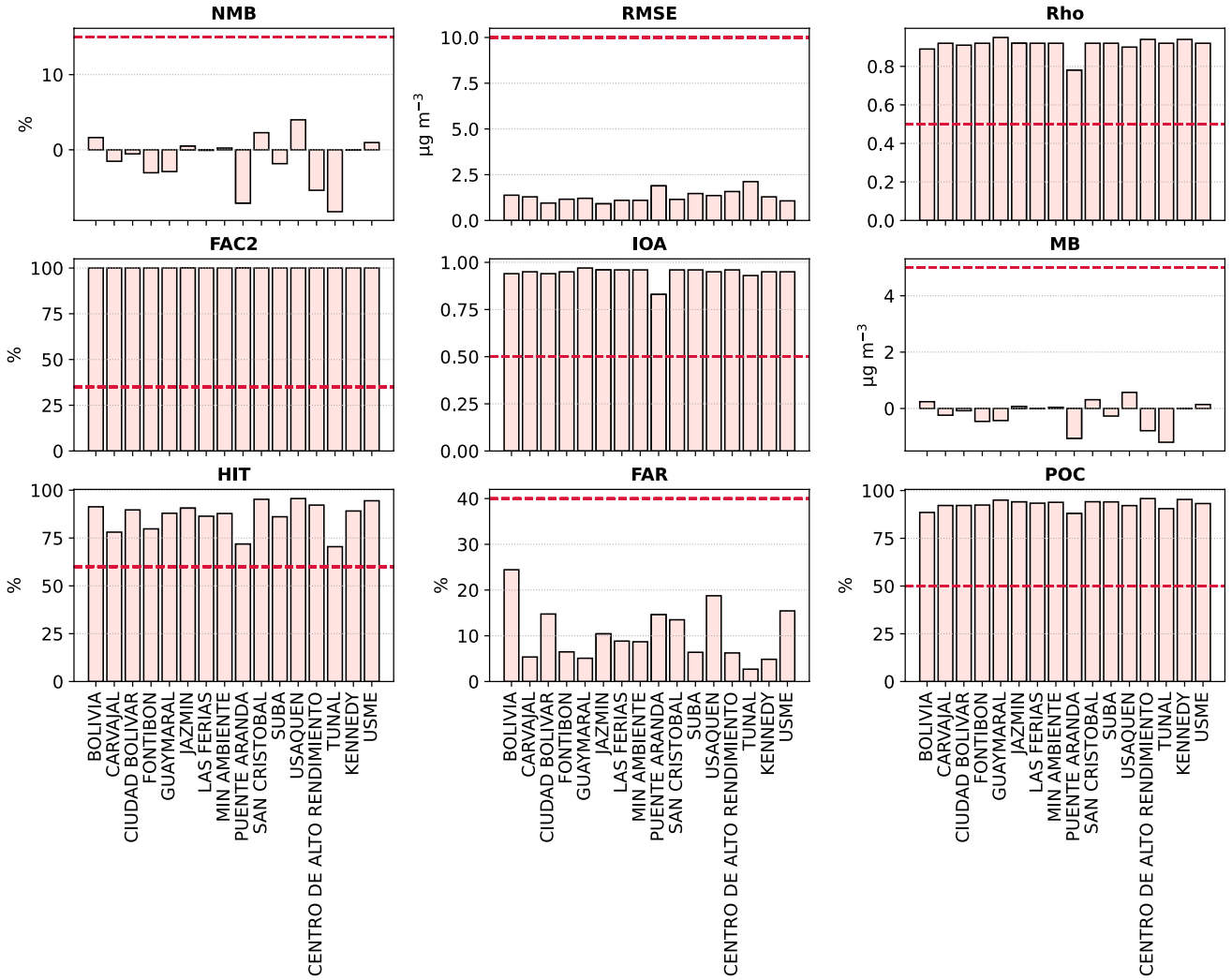


Fig. 13 Statistical parameters calculated from daily data for the 16 air quality monitoring stations in Bogotá, selected for the neural network validation. This validation uses the daily values calculated as the maximum 8-hour running mean of ozone for both the model and the ground station for 182 days (see "Methods"). Each panel shows an evaluation parameter, and the red lines represent the performance benchmarks

capture local ozone variability in data-rich urban environments, thereby complementing physically based models in practical air-quality management applications. This predictive capability, while not flawless, represents a significant step toward proactive air quality management.

To illustrate this proposed integration, Fig. 14 presents a simplified workflow of how the model could function within a real-time operational EWS. The system begins with continuous acquisition of air quality and meteorological data, which is processed and fed into the model to produce hourly ozone forecasts. When forecasted concentrations exceed predefined health-based thresholds (e.g., WHO or national standards), an internal alert is automatically triggered within the relevant local environmental agency. This alert does not immediately result in public notification; rather, it initiates a verification process in which experts assess the evolving pollution dynamics and regional transport patterns. Only after this assessment would the appropriate type and scale of public warning be determined and disseminated; minimizing unnecessary alerts while ensuring timely responses when genuine risks are present (Casallas et al. 2022).

4 Conclusions

Our analysis reveals a clear and sustained upward trend in ozone concentrations across Bogotá from 2013 to 2023, with particularly steep increases in urbanized and socioeconomically vulnerable areas such as Suba, Kennedy, and Ciudad Bolívar. These localities not only have high population densities and heavy traffic volumes but also limited green infrastructure and exposure to prolonged sunlight, conditions that are highly conducive to photochemical ozone formation. The magnitude of the increase in these areas

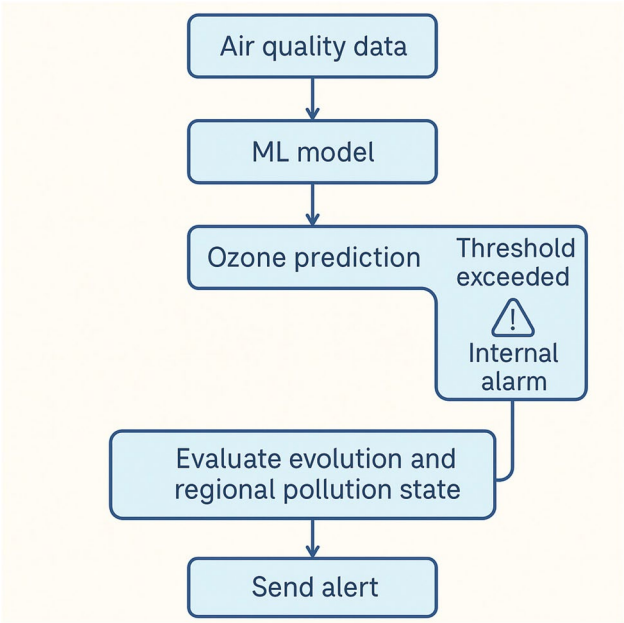


Fig. 14 Workflow for using the CNN–BiLSTM model in a real-time ozone early warning system. The model takes air quality and weather data to predict ozone levels. If predicted values exceed health-based thresholds, an internal alert is triggered so experts can review the situation and decide whether to issue a public warning

suggests that local emission patterns and urban structure play a dominant role in shaping ozone dynamics.

Importantly, this trend is not homogeneous across the city. While some stations show marked increases, others (often located in areas with higher NO_x emissions or better atmospheric ventilation) exhibit stable or even slightly decreasing trends. This spatial heterogeneity indicates that ozone formation in Bogotá results from a complex interaction between local and regional drivers, including vehicular and industrial emissions, NO titration effects, solar radiation, and seasonal (regional) events like biomass burning. Thus, the rise in ozone levels cannot be attributed solely to broader regional or meteorological patterns. These findings emphasize the need for area-specific strategies that address the particular emission profiles and environmental vulnerabilities of different parts of the city.

From a health perspective, our analysis estimates that 18.3% of all deaths among individuals aged 25 and older in Bogotá between 2013 and 2023 are attributable to long-term exposure to ozone. The burden is not evenly distributed: older adults (particularly in localities like Ciudad Bolívar, Teusaquillo, Puente Aranda, and Kennedy) face a disproportionately higher risk. This group may be more vulnerable due to a combination of biological susceptibility, socioeconomic conditions, limited access to healthcare, and higher exposure in densely populated or poorly ventilated neighborhoods.

This spatial and demographic inequality highlights an emerging and uneven public health crisis linked to air pollution. Without targeted interventions, these impacts are likely to intensify as ozone levels continue to rise, especially in areas where urban growth, traffic density, and social vulnerability intersect. These findings underscore the urgent need not only for localized public health and environmental strategies, but also for the development of predictive models and EWSs that can identify high-risk periods and guide timely, data-informed responses to protect the most affected populations.

To this end, Bogotá's air quality and public health management should prioritize: (i) the deployment of localized emissions-reduction strategies, such as retrofitting public transport fleets, enhancing VOC measurements (the city lags them), and NO_x control in industrial zones, and enforcing low-emission corridors; (ii) the expansion of green buffers and urban vegetation in ozone-vulnerable neighborhoods; and (iii) the establishment of community-level risk communication programs tied to early-warning ozone forecasts. These measures can work synergistically to reduce exposure and improve resilience in the most burdened areas.

As a first step toward an EWS, we developed a machine learning model using a CNN-BiLSTM architecture, trained on meteorological and air quality data. The model showed strong performance in capturing daily ozone trends, diurnal cycles, and overall variability, with correlation values reaching up to 0.87 in some stations. Its ability to learn temporal patterns makes it a promising tool for operational ozone forecasting, especially in areas with consistent data and moderate emission variability.

However, the model tends to underestimate sudden ozone peaks, particularly in complex urban settings like Puente Aranda. This limitation likely stems from the absence of key precursors, such as NMHCs, and challenges in representing abrupt emission or meteorological shifts. To improve peak detection and reliability under extreme conditions, future work could explore specialized loss functions, stochastic modeling, and integration of additional variables. Despite these challenges, the model offers a solid foundation for real-time alerts and public health risk reduction in Bogotá.

In summary, this work leads to three key conclusions:

Ozone levels are rising across Bogotá in ways that are shaped by both local emissions and broader (regional) meteorological conditions. If left unaddressed, these increases could significantly degrade air quality and public health over the coming years.

The health burden of ozone exposure is already substantial and unequally distributed, pointing to an urgent need for localized mitigation policies, particularly in the most affected districts.

Machine learning offers a powerful tool for anticipatory environmental management, and with continued development, can form the backbone of real-time ozone warning systems that inform both authorities and at-risk populations.

Given the convergence of urban expansion, meteorological variability, and rising ozone levels, Bogotá faces a critical window for intervention. This study offers evidence-based tools and insights to support proactive, data-driven air quality strategies that protect health and improve urban resilience in the years ahead.

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Data availability In-situ station data are available for download from IDEAM (2023). Population data can be obtained from DANE (2024), while health-related information is accessible through DANE (2019, 2025). All the scripts used in this research can be found in https://github.com/jdanbq/Ozone-Trends-and-Mortality-Risk.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Clinical trial number Not applicable.

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